

TYRE DEGRADATION MODELLING IN FORMULA 1

A Data-Driven Machine Learning Pipeline



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ME228 Course Project

THE CORE CHALLENGE

Why It Matters

Pit stop timing, compound choice & stint length all depend on accurate tyre wear models. With an ever evolving competition, such analysis is really important to strategize

Isolating Signal

Separating true degradation from race events is the central challenge

Noisy Lap Times

Slow laps $\neq$ tyre wear



TWO-STAGE PIPELINE

STAGE 1

CLEAN LAP CLASSIFIER

MLP Neural Network

V1: Purely Gap Based Labels

GapAhead / GapBehind thresholds

V2: Telemetry Labels

240 Hz speed, throttle, brake signals

Output: $CleanProb \in [0,1]$ per lap



STAGE 2

DEGRADATION LEARNING

Ridge Regression

Gradient Boosting

Random Forest

MLP Regressor

DATA COLLECTION & PREPROCESSING

22

2024 Races

6

Telemetry Races

240Hz

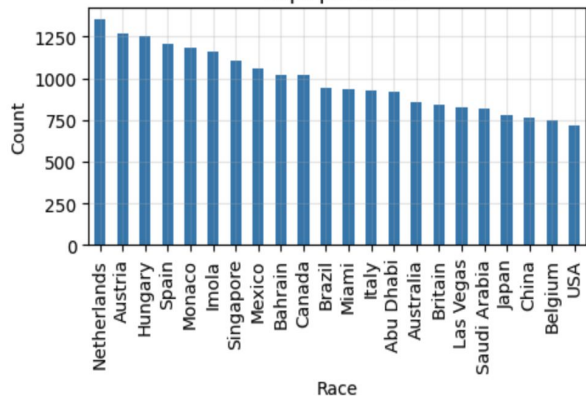
Car Telemetry

$$\Delta T_i = t_i - \min(t_{\square})$$

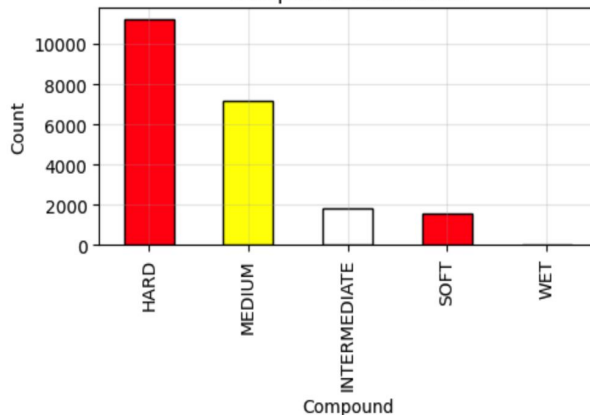
Regression Target

Fig 1 — Dataset Overview (2024, 22 races)

Laps per Race



Compound Distribution



Filtering Steps

IsAccurate = True only

Pit laps removed

Laps > 200s dropped

3 σ outlier removal per compound

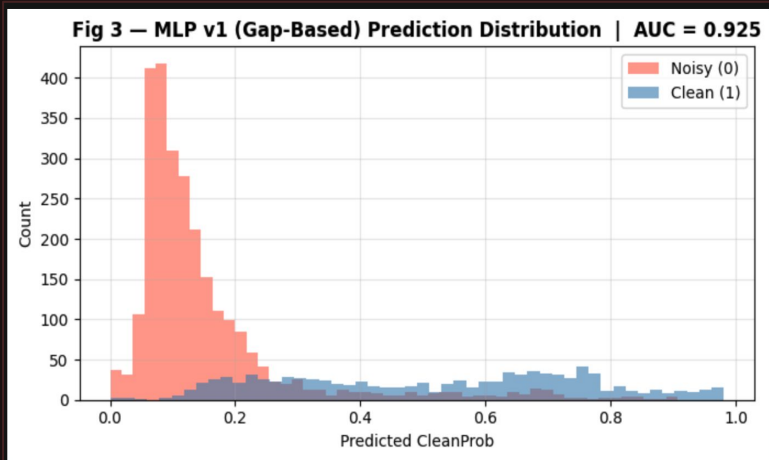
CLEAN LAP CLASSIFICATION

V1 — GAP-BASED

Clean if GapAhead & GapBehind > 3.45s

Noisy if either gap < 2.55s

Borderline laps discarded

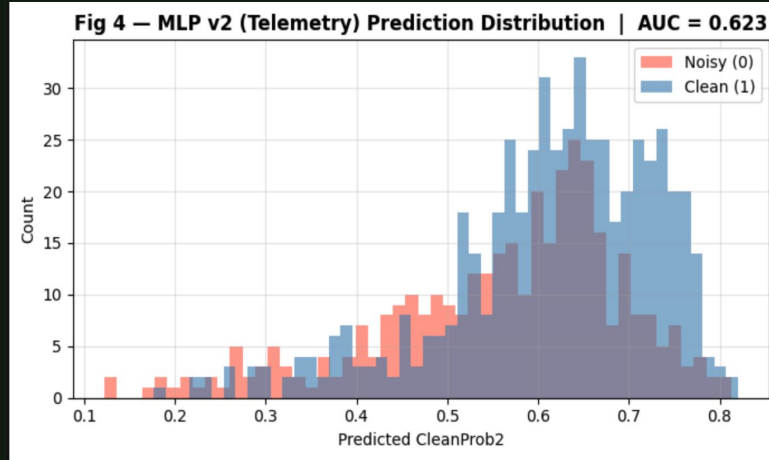


V2 — TELEMETRY

Speed std · Throttle std · Full Throttle%

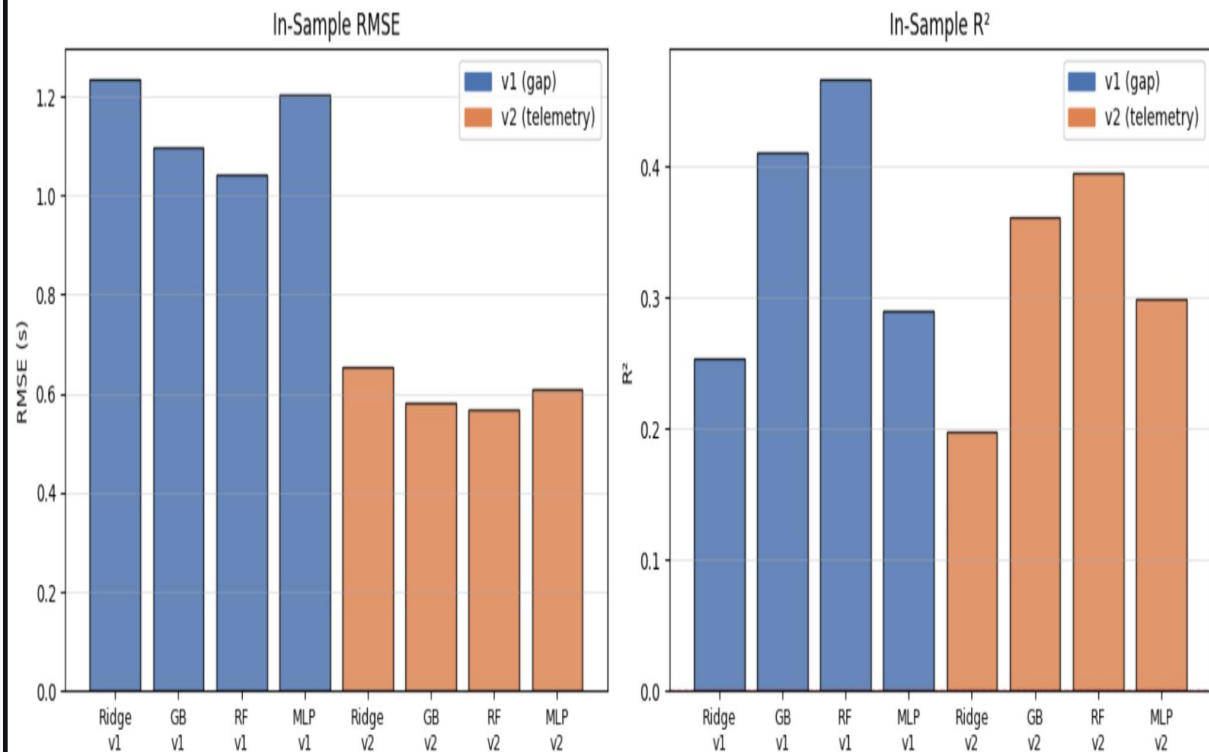
Brake events · Speed drop events

Z-scored within stint → composite score



IN-SAMPLE PERFORMANCE — 2024 SEASON

Fig 8 — In-Sample Model Comparison (all 8)



BEST PERFORMERS

RF (V2)

RMSE 0.566 R² 0.395

GB (V2)

RMSE 0.582 R² 0.362

MLP Reg (V2)

RMSE 0.610 R² 0.299

Ridge (V2)

RMSE 0.650 R² 0.197

OUT-OF-SAMPLE: BAHRAIN 2023

Tested on unseen race from a different season

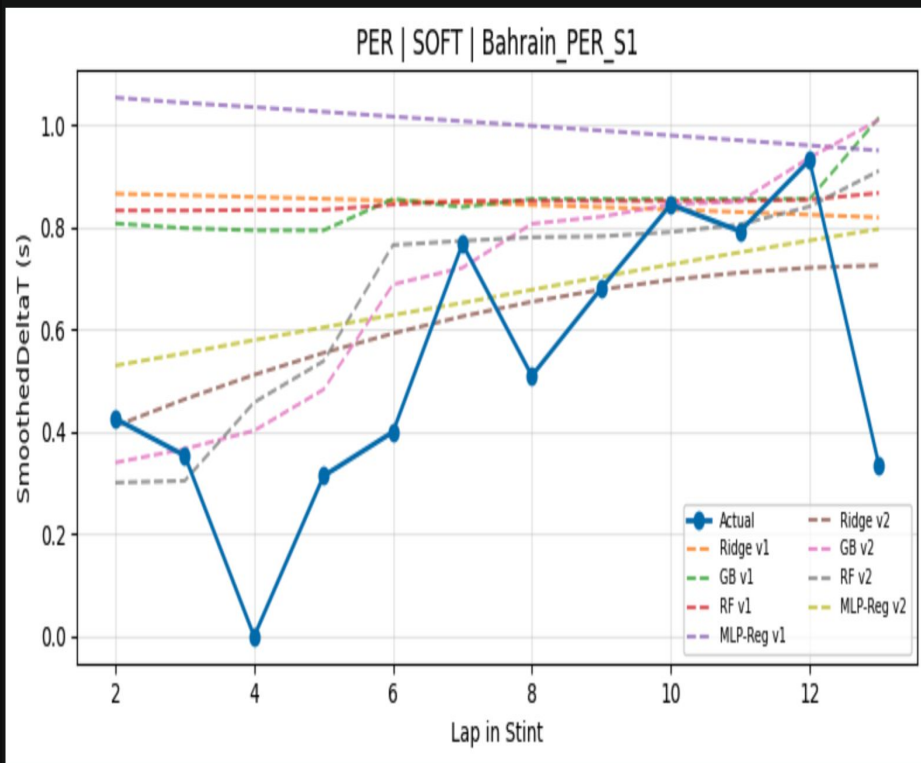
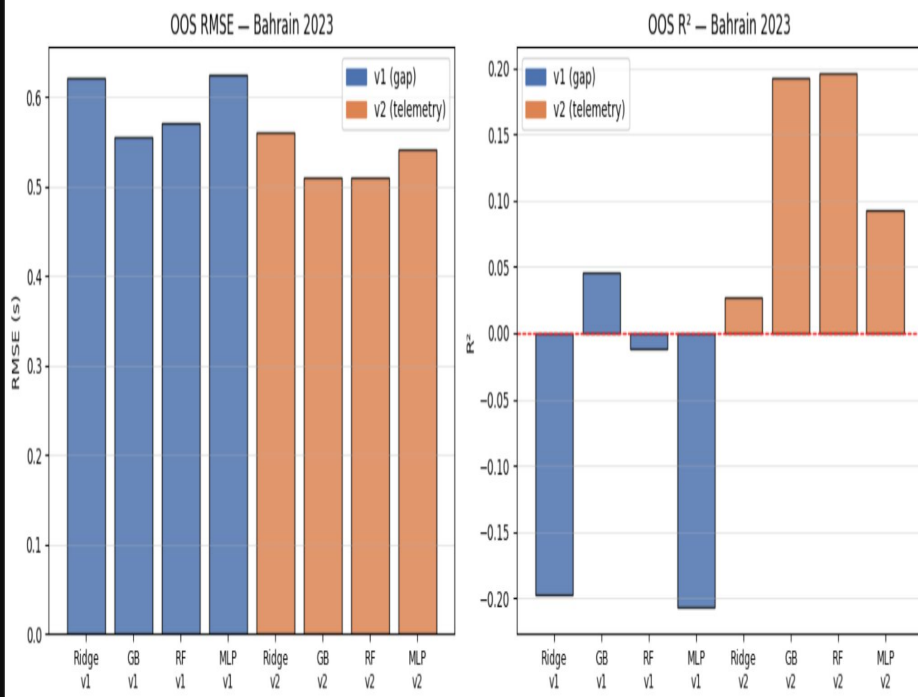


Fig 9 – Out-of-Sample Comparison (Bahrain 2023, all 8)



CONCLUSIONS & FUTURE WORK

Gradient Boosting and Random Forest with telemetry labels (V2) give the most reliable tyre degradation predictions.

FINDINGS

- V2 (telemetry) > V1 (gap) labels
- Race encoding is essential
- GB & RF best overall performers
- MLP needs more tuning

LIMITATIONS

- Cross-year drift persists
- Race encoding $\neq$ track physics
- Telemetry = high compute cost
- Fuel load confounds early stints

FUTURE WORK

- Physics-based track features
- Fuel load correction
- Full 22-race telemetry
- Live race strategy integration